Axel Gomez

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Axel Gomez AxelGomez16 Web profile
Queretaro, Mexico — Open to Relocation

Professional Summary

Algorithms Engineer with **4+ years in automotive R&D**, developing **C++/Python** software for perception, parallel algorithm orchestration, and sensor validation. Migrates legacy code, integrates components through eCAL, and builds evaluation tools that support release decisions. Also owns development, customer support, and feature delivery for a subscription publishing product.

Professional Experience

Aumovio (formerly Continental R&D) June 2022 - Present
Algorithms Engineer *Queretaro, Mexico*

- **Radar Model Validation:** Proposed a **Python** pipeline, then refactored and migrated validation code with **GitHub Actions**, reducing a multi-engineer workflow from approximately **five working days to 50 minutes** for General Motors.
- **Algorithm Orchestration:** Migrated legacy **C++** orchestration to new camera/radar parking-system requirements, supporting parallel algorithm flows and per-frame failure handling.
- **Team Enablement & Code Quality:** Set up Linux environments, trained teammates on internal tools, and helped resolve QAC findings across algorithm components under AUTOSAR/MISRA.
- **LiDAR Perception & Evaluation:** Developed **C++** algorithm and point-cloud visualization features using **eCAL**; built Python/SQLite/Dash/Plotly evaluation tools to make performance data available across teams.
- **Perception Integration:** Established two-week checks of combined algorithm performance to inform release decisions; automated commit/pull-request checks to surface detection and bounding-box-query regressions.

Postealo - Independent Software Product December 2024 - Present
Developer & Maintainer (alongside full-time employment) *Queretaro, Mexico*

- Built a **Python** product connecting bot intake, **LLM-assisted** editorial preparation, and immediate/scheduled publishing across a client's website, Facebook, Instagram, X, and WhatsApp.
- Maintain the product for a paying news organization under a **monthly subscription**, translating requirements into features and providing ongoing support.

Continental Talent Entry Program (CTEP) March 2022 - June 2022
Systems Engineer Trainee *Queretaro, Mexico*

- Completed requirements, development, and testing work on a door-lock and window-control vehicle prototype before transitioning to algorithms engineering.

Technical Skills

Programming: Python, C++, C, MATLAB
Build & CI: Git, GitHub Actions, Jenkins, Linux, CMake, Docker
Systems & Quality: Bash, eCAL, parallel orchestration, QAC, AUTOSAR/MISRA, regression checks
Data & Applications: SQLite, Dash, Plotly, NumPy, Pandas, OpenCV
Spoken Languages: English, Spanish

Education

Master of Science in Computer Science (Artificial Intelligence) Expected July 2027
Autonomous University of Queretaro *Online, 2025 - Present*

Bachelor of Science in Engineering Physics 2017 - 2021
Autonomous University of Queretaro *Queretaro, Mexico*